# AMITASH NANDA

San Diego, CA (USA) | (858) 319-5516 | ananda@ucsd.edu | linkedin.com/in/amitash01 | amitashnanda.github.io

## EDUCATION

**University of California San Diego**, San Diego (CA), USA — September 2023 – June 2027
Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering (Machine Learning and Data Science) — GPA: 3.5
Research: Distributed ML, Generative AI, Computer Vision, DNN/LLM Optimization, AI/ML Systems, Applied Data Science

**University of California San Diego**, San Diego (CA), USA — September 2021 – June 2023
Master of Science (M.S.) in Electrical and Computer Engineering (Intelligent Systems, Robotics and Control) [**Thesis**] — GPA: 3.5

**Odisha University of Technology and Research**, Bhubaneswar (OD), India — August 2015 – May 2019
Bachelor of Technology (B.Tech.) in Instrumentation and Electronics Engineering — GPA: 4.0

## TECHNICAL SKILLS

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|---|---|
| **Languages/Databases** | Python, C++/CUDA, JavaScript, Bash/Shell, SQL (MySQL, PostgreSQL), GraphQL, MongoDB, Redis |
| **Frameworks** | PyTorch, TensorFlow, Keras, Hugging Face Transformers, DeepSpeed, Spark, ONNX, YOLO, Flask |
| **Libraries** | Scikit-learn, Pandas, NumPy, Dask, OpenCV, NLTK, OpenMP, TensorRT, cuDNN, SHAP, Captum |
| **Tools** | Anaconda, Jupyter, LaTeX, Git/GitHub, Jenkins, Docker, Kubernetes, Slurm, Grafana, MLflow, MLPerf |
| **Cloud/Technologies** | AWS (SageMaker, EC2, Lambda), Azure ML, GCP, HPC Systems (NERSC Perlmutter), LangChain |

## EXPERIENCE

**Graduate Research Assistant | Lawrence Berkeley National Laboratory**, Berkeley (CA), USA — June 2025 – September 2025
- **Data and Analytics team:** Led the development of a foundation AI model for protein understanding and design on HPC.
- Engineered a novel guided-generation framework integrating **ESM-3** and custom scoring function for protein stability.

**Graduate Research Assistant | Lawrence Berkeley National Laboratory**, Berkeley (CA), USA — June 2024 – September 2024
- Expanded load-balancing algorithms in **AMReX**, parallelized a brute-force approach, developed a hybrid **SFC-Knapsack** and improved SFC bisection strategy using **painter's**, achieved faster run-time, and **99.8–99.9%** balanced efficiency at scale.

**Teaching Assistant | CSE, Data Science, ECE (UCSD)**, San Diego (CA), USA — March 2024 – Present
- Managed **60** students across **six** groups for the Data Science Capstone Project, guided them through **ML model optimization**, **NLP**, **Gen-AI**, and **Graph ML**. Assisted as a TA in the Advanced Data Structures course (**CSE 100**).
- Facilitated discussions, office hours, and assessments, supported group projects and provided guidance for code checkpoints.

**Graduate Student Researcher | Boolean Lab (UCSD)**, San Diego (CA), USA — September 2021 – Present
- **Computational Models:** Analyzing large-scale biological datasets to identify **Boolean** relationships between genes and developing computational methods to accelerate the use of AI in pathology and drug discovery.
- **AI Integration and Model Optimization:** Building multimodal approaches, foundation models, and fine-tuning strategies for vision and language-related tasks to generate biological hypotheses and identify candidate biomarkers.

**Software Engineer Intern | Teradata (Optimization Group)**, San Diego (CA), USA — June 2023 – September 2023
- **Query Optimization:** Leveraged Primary Index (PI) for efficient local aggregation and join capabilities to enhance Teradata's Object File Storage (OFS) system and achieved **20%** query cost reduction. Optimized data scenarios for **1B** rows.
- **Framework:** Developed an automated framework for data and query generation, simulated good and bad cases for **DBQL** performance benchmarking. Generated **512** objects with **700M** rows (good case) and **233** objects with **48M** rows (bad case).

**Research Intern | Teradata (Technology Innovation Office)**, San Diego (CA), USA — June 2022 – September 2022
- **Performance Optimization:** Integrated predictive models to forecast platform configuration resource usage, analyzed telemetry data from the **Teradata Telemetry Collection Agent** (TCA), and improved operational efficiency.
- **Automation:** Designed and implemented an end-to-end automated pipeline, integrated data retrieval from TCA using **REST** API, performed data preprocessing, built predictive models, and streamlined platform monitoring and forecasting.

**Software Engineer II | Accenture (Application Engineering R&D)**, Bengaluru (KA), India — June 2019 – August 2021
- **Application Development:** Designed and built a novel robot software testing application (**chaosRobo**) utilizing chaos engineering principles to simulate real-world scenarios. Led the **first-phase** prototype development team.
- **Testing Framework:** Contributed to backend **GUI** design, integrated **Gazebo-ROS** functionalities, and successfully deployed the system on **AWS S3** and **RoboMaker**, enhancing robot simulation and testing efficiency.
- **Performance Optimization:** Implemented machine learning-based analytics in chaosRobo to assess system robustness, reducing failure detection time by **30%** and improving overall testing accuracy by **20–25%**.
- **Bedrock Automation Asset Development:** Automated end-to-end user stories for **Airbus** PPR scenarios using image-based and text-based recognition. Devised a new testing methodology and integrated it into the **3DEXPERIENCE** software.

## PROJECTS

**ESM-3 Guided-Generation Based Protein Engineering | NERSC (LBNL) [GitHub] [Package]** June 2025 – Present

- Developed a guided-generation framework integrating **ESM-3** generative model with **FoldX** scoring function (**ddG** scores).
- Enhanced framework capabilities to refine sequences for thermostability and binding affinity, reducing time for each generation step by **75–80%** and achieving **8x** faster performance leveraging **HPC-AI** workflow. Published a **PyPI** package.

**Dynamic Load Balancing Algorithms in AMReX | NERSC (LBNL) [GitHub]** June 2024 – March 2025

- Introduced a novel hybrid load-balancing algorithm combining SFC and Knapsack, improved SFC bisection and Knapsack using **painter's** algorithm, optimized dynamic load-balancing in AMReX framework on NERSC Perlmutter supercomputer.
- Statistically analyzed the efficiency and runtime of these algorithms for sizes up to **512** ranks. Knapsack combined with painter's achieved highest efficiency ranging **99.8–99.9%**, while SFC with painter's algorithm demonstrated faster runtime.

**CHAI-KTQ: Framework for Scalable LLMs & Efficient Inference** | Boolean Lab [**GitHub**] August 2024 – Present

- CHAI-Quant performs sensitivity-aware mixed-precision quantization on clustered attention heads, reducing KV cache size by up to **57.8%**. CHAI-Target performs sensitivity-aware fine-tuning improving accuracy **22–24%** over baseline CHAI.
- CHAI-KD implements an efficient teacher-student knowledge transfer, achieving inference speed of **3x** (**3,000 inf/sec** ).

**CPTQuant: Mixed Precision Post-Training Quantization for LLMs** | Boolean Lab [**GitHub**] May 2024 – Present

- **CMPQ** adjusts precision using canonical correlation analysis across layers. **PMPQ** applies layer-wise precision optimization based on sparsity sensitivity. **TDMPQ** leverages Taylor decomposition to adapt precision to input perturbations.
- **CPTQuant** achieved up to **4x** compression and a **2x** efficiency over Hugging Face **FP16**. **PMPQ** delivered **11%** higher compression for classification tasks, while **TDMPQ** achieved a **30%** greater compression ratio for language modeling tasks.

**Communication Efficient Asynchronous Peer-to-Peer Federated LLMs** | Boolean Lab [**GitHub**] December 2023 – April 2024

- Developed a secure, efficient, and privacy-preserving federated learning framework in a decentralized setting, achieving **5x** lower latency and **13%** higher accuracy in serverless environments.
- Observed a **76%** reduction in information passing time for asynchronous methods, while PageRank proved most effective at removing anomalous nodes, boosting the global federated LLM model's performance.

**Expression Gradient of Cancer Suppressor Gene using Vision-AI** | Boolean Lab [**GitHub**] May 2022 – May 2023

- Built a novel colon gland instance segmentation model using Mask R-CNN and YOLOv8 to diagnose colon cancer. Mask R-CNN achieved F1/IoU scores of **0.63/0.46** (glands) and **0.59/0.42** (crypts). Annotated first public U-shaped colon dataset.
- Demonstrated that YOLOv8 outperformed Mask R-CNN with mAP50 scores of **0.937** (glands) and **0.748** (crypts). Predicted **5,000+** U-shaped glands across **25** slides, identifying differentially expressed genes along crypts.

## PUBLICATIONS

- **Nanda, A.**, Chowdhury, M.K.H., Ross, H., and Gott, K. (2025). Exploring Dynamic Load Balancing Algorithms for Block-Structured Mesh-and-Particle Simulations in AMReX. In *Proceedings of the ACM PEARC*. **DOI: 10.1145/3708035.3736022**
- **Nanda, A.**, Balija, S.B., and Sahoo, D. (2025). CHAI-KTQ: A Novel Framework for Scalable LLMs and Efficient Inference. Manuscript in preparation.
- Behroozikhah, M., **Nanda, A.**, Khandelwal, S., and Sahoo, D. (2025). Expression Gradient of Cancer Suppressor Gene Found in Colon Crypt using Vision-AI. Manuscript in preparation.
- **Nanda, A.**, Haque, H.M.Z., Zhou, X., and Sahoo, D. (2025). GeneST: A Deep Learning Framework to Identify Spatially Distributed Genes for Translational Health Impact. Manuscript under review.
- **Nanda, A.**, Balija, S.B., and Sahoo, D. (2025). FedNAMs: Performing Interpretability Analysis in Federated Learning Context. Preprint. arXiv. **DOI: 10.48550/arXiv.2506.17466**
- **Nanda, A.**, Balija, S.B., and Sahoo, D. (2024). CPTQuant: Novel Mixed-Precision Post-Training Quantization Techniques for Large Language Models. Preprint. arXiv. **DOI: 10.48550/arXiv.2412.03599**
- Balija, S.B., **Nanda, A.**, and Sahoo, D. (2024). Building Communication Efficient Asynchronous Peer-to-Peer Federated LLMs with Blockchain. In *Proceedings of the AAAI Spring Symposium Series*. **DOI: 10.1609/aaaiss.v3i1.31212**
- **Nanda, A.**, Sahoo, D., and Lin, B. (2023). Novel Vision-AI Techniques for Morphological Discovery in Systems Biology. Master's Thesis, University of California San Diego, *Open Access Publication*. **URL: escholarship.org/uc/item/0xh5w6wb**
- Dang, D., **Nanda, A.**, Sahoo, D., and Lin, B. (2023). NeuCASL: From Logic Design to System Simulation of Neuromorphic Engines. Preprint. arXiv. **DOI: 10.48550/arXiv.2208.03500**

## RECOGNITION / OPEN-SOURCE CONTRIBUTION

- MLCommons ( **MLPerf Inference v5.1**) – Benchmarked **ResNet-50** with Apple M1 Pro (10-core CPU, 16-core GPU) using ONNX runtime v1.19.2 with Apple's CoreML/ANE across Offline, SingleStream, and MultiStream. [**Website**] [**GitHub**]
- Reviewer for AAAI 2026, ICLR 2025-2026, Poster Judge for Academic Data Science Alliance 2023 (UTSA).
- Selected for **ACM PEARC** 2025 student program, served as Poster Judge, facilitated Nvidia's workshop as a lead volunteer.